ALPhANOV
Optics & Lasers Technology Center

ALPHANOV

Electronic Components Evaluation

CASCADE 2026

Benoît Morin

www.alphanov.com

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MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR, DE LA RECHERCHE ET DE L'ESPACE

RÉGION Nouvelle-Aquitaine

ALPHA RLH

cnrs

université de BORDEAUX

cea

CRT

ISO 9001:2015 BUREAU VERITAS Certification

Property of ALPhANOV

ALPhANOV - Introduction

ALPhANOV is an optics and lasers technology center based in Nouvelle-Aquitaine (France)

2007
Foundation

7-8.0 M€
Turnover

25 Start-Up
supported over the past 15 years

120+
Employees

20+
Running collaborative projects

2 500+
Customers/partners accumulated

ALPhANOV develops solutions for:

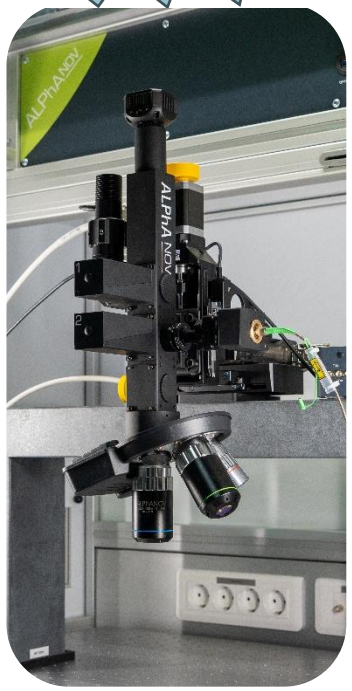
- Health
- Defense, Aeronautics and Space
- Industry
- Luxury
- Quantum applications
- Lasers
- Cybersecurity

5 Business Units…

Laser processing

Laser sources and components

Photonic systems

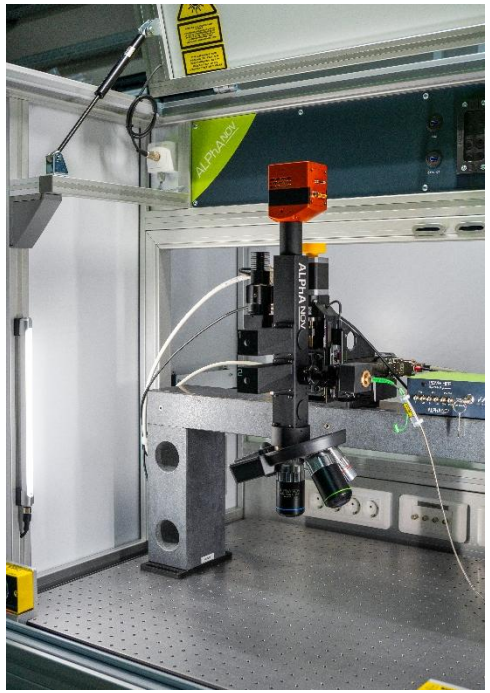


Electronic Components Evaluation

Training Center

ECE Department - Introduction

- ECE develops, manufactures and sells **high-end solutions** for **electronic components evaluation** (LFI, PE, etc…)
- In **IC testing** domain **since 2009**
- Our **contribution & role** in this domain is to bring **expertise** and **innovations** in **optics & laser**
- Types of customers which use our solutions :
  - Labs & universities
  - Governmental organizations
  - Certification companies
  - Chip manufacturers
  - Smart devices makers

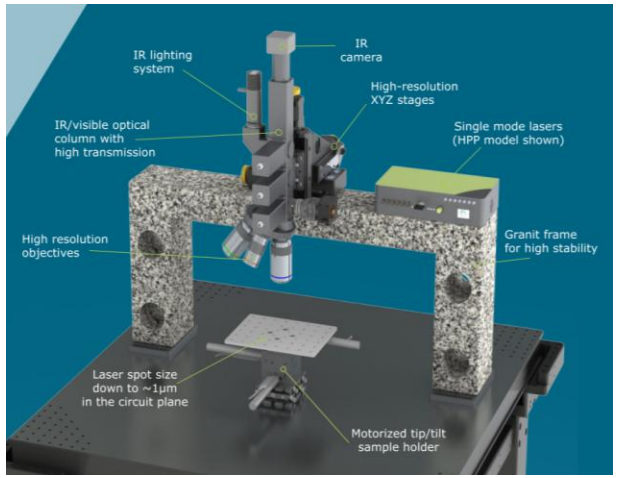


- **Installed base** : **>80 systems** worldwide (equipment subject to export control)
- Our **products & services** are recognized by the community for their **performances**, **quality**, and **reliability**
- A strong **partnership** with eShard which offers complementary solutions for EM attacks and provides complete S/W environment

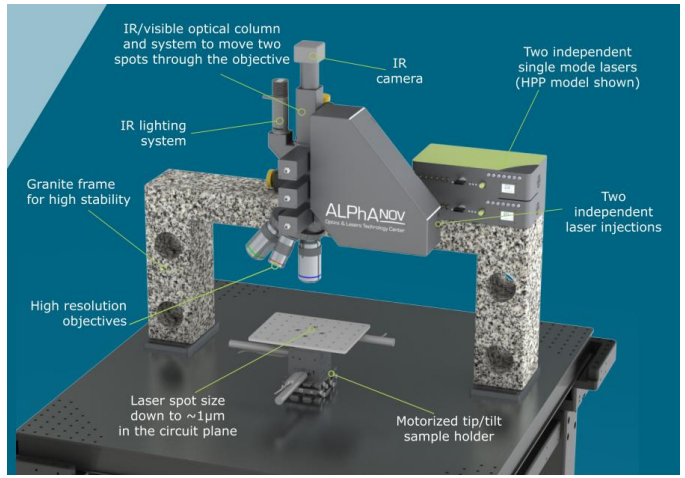
Overview – LMS offers

LFI STATIONS

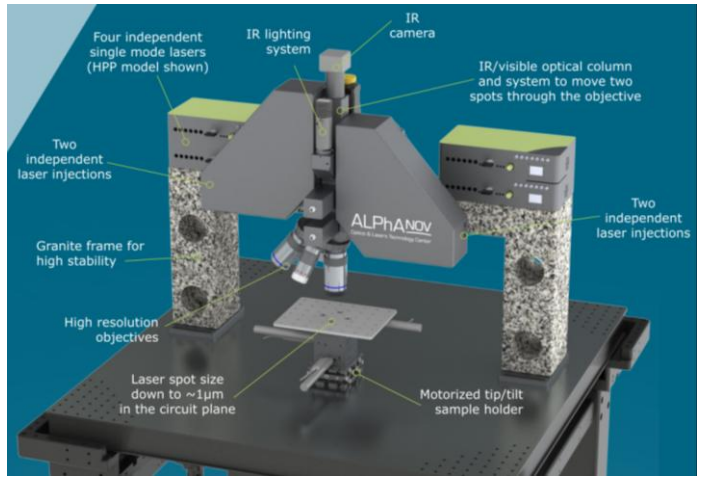
S-LMS
Single Laser Microscope Station



D-LMS
Double Laser Microscope Station



Q-LMS
Quadruple Laser Microscope Station



OPTIONS

Photon Emission Imaging	Photon Emission Counting	Thermal Laser Stimulation	Galvo Scanning

SOURCES

Nanosecond Laser	Picosecond Laser	Femtosecond Laser

Other options/services: esDynamic for Chip Security **– Demo Campaign - Chip Decapsulation Campaign – H/W Custom Requests**

Lasers sources & absorption

SPA
Single Photon Absorption

TPA
Two Photon Absorption

PDM
Nanosecond laser source

Wavelength: 808, 980, 1064, 1420nm

Pulse width: ~1ns (tunable)

Jitter: <20ps

LFI Pico
Picosecond laser source

Wavelength: 1064nm

Pulse width: ~30ps

Jitter: <100ps

LFI Femto
Femtosecond laser source

Wavelength: 1560nm

Pulse width:~500fs

Jitter: <25ns

**PDM: Pulse on Demand Module*

Thank you for your attention

ALPhANOV
Centre Technologique Optique et Lasers

Benoît Morin
Sales

+33 6 50 78 05 45